

## BHAGAMANDALA HOUSE — COMPLETE CONSTRUCTION PLAN (v4)

Location: Bhagamandala–Talakaveri Road (Curve), Kodagu, Karnataka

3 km from Talakaveri Temple, Left side (heading to temple)

Plot: 60 ft (E-W) × 50 ft (N-S) | East-Facing | Highway on East

Date: April 2026

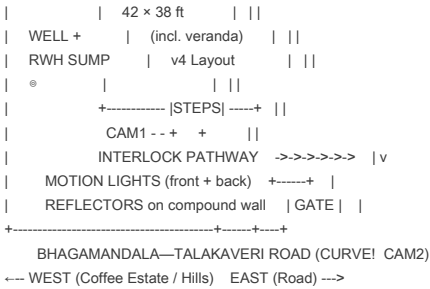
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## 1. SITE LAYOUT PLAN (60 ft × 50 ft Plot)

NORTH  
←----- 60 ft ----->

+-----+  
| 7' SOLID COMPOUND WALL (crash protection from hwy) | ^  
+-----+ +-----+ ||  
| OPEN | COFFEE DRYING | \* Tulsi ||  
| GARAGE | YARD 30×20 ft | Vrindavan ||  
| 15×25 ft | PCC + Retractable | (NE) ||  
| 2 Cars | Polycarbonate | ||  
| | Roof | CAM3 ||  
| CAM4 | | ||  
+-----+ +-----+ ||  
+-----+ ||  
| TOOL | GARDEN / LAWN ||  
| SHED | 50 ft ||  
| 10×8 ft | FF SITOUT (above) ||  
+-----+ 26×8 — West views ||  
+-----+ ||  
| SEPTIC | | ||  
| TANK | MAIN HOUSE | ||  
| (NW) | G+1 BUILDING | ||



Site Layout Notes:

- Setbacks: Front (East) 5 ft, Rear (West) 3 ft, North 5 ft, South 3 ft
- Main Gate: SE area, 12 ft vehicle + 4 ft pedestrian, SET BACK from curve apex
- Compound wall: 7 ft solid laterite on road side (crash protection from highway curve); 6 ft solid on other sides. NO MS railing on top — solid throughout for security (isolated house).
- CCTV: 4 cameras — CAM1 (gate), CAM2 (road/curve), CAM3 (back), CAM4 (parking)
- Motion sensor LED floods: Front (gate area) + Back (estate side)
- Reflectors: On compound wall facing highway (night visibility for vehicles)
- Tulsi Vrindavan: NE corner (Vastu — most auspicious for sacred plants)
- Interlock tiles: Herringbone pattern grey for all pathways and driveway

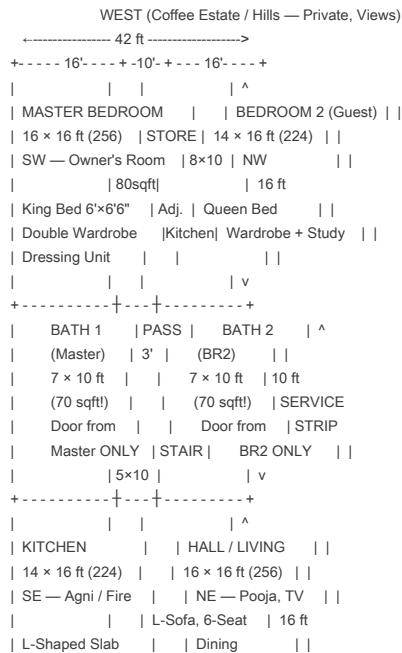
2. GROUND FLOOR PLAN (v4 — Optimized)

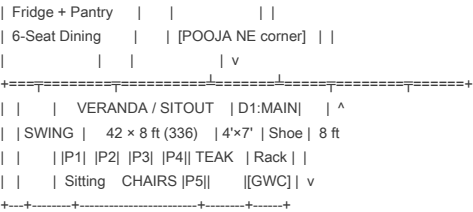
Building: 42 ft (N-S) × 38 ft (E-W) total footprint

Interior: 42 ft (N-S) × 30 ft (E-W) behind veranda

Veranda: 42 ft × 8 ft (East, road side)

Plinth: 4 ft above ground (mandatory — 5,000mm rainfall)





KEY v4 FEATURES:

- BACK-TO-BACK BATH CORE: Bath1 + Bath2 share ONE plumbing stack
- BEDROOMS: Full 16x16 and 14x16 RECTANGLES (no L-shapes!)
- BATHS: 7x10 = 70 sqft each — separate shower, WC, basin, geyser
- Kitchen ADJACENT to Store — shared wall with door
- Passage 3 ft wide connects north row to south row
- 5 Chadara Kamba wooden pillars (wider building needs 5)
- GWC 4x4 in veranda NE corner

Ground Floor Room Schedule (v4)

| Room                | Size       | Area     | Position (Vastu)     | Notes                                                        |
|---------------------|------------|----------|----------------------|--------------------------------------------------------------|
| HALL / LIVING (NE)  | 16 × 16 ft | 256 sqft | NE — ಈಶಾನ್ಯ          | L-sofa, TV unit, 6-seat dining, Pooja cabinet NE corner      |
| MASTER BEDROOM (SW) | 16 × 16 ft | 256 sqft | SW — ನೈರುತ್ಯ         | King bed, double wardrobe, dressing unit. FULL RECTANGLE     |
| BEDROOM 2 (NW)      | 14 × 16 ft | 224 sqft | NW — ಪಾಯುವು          | Queen bed, wardrobe, study desk. FULL RECTANGLE              |
| KITCHEN (SE)        | 14 × 16 ft | 224 sqft | SE — ಅಗ್ನಿ           | L-slab, fridge, 6-seat dining, pantry. Adj. to Store!        |
| BATH 1 (Master)     | 7 × 10 ft  | 70 sqft  | Center service strip | Shower 4x3, WC, basin counter, geyser. Door from Master ONLY |
| BATH 2 (BR2)        | 7 × 10 ft  | 70 sqft  | Center service strip | Shower 4x3, WC, basin counter, geyser. Door from BR2 ONLY    |
| STORE               | 8 × 10 ft  | 80 sqft  | West, adj. kitchen   | Floor-to-ceiling shelving. Kitchen access via internal door  |
| STAIRCASE           | 5 × 10 ft  | 50 sqft  | Center-East          | U-turn granite steps, under-stair storage                    |
| PASSAGE             | 3 × 10 ft  | 30 sqft  | Center               | Connects north & south rows through service strip            |
| GUEST WC            | 4 × 4 ft   | 16 sqft  | NE veranda corner    | For visitors. WC + wash basin. No house entry needed         |
| VERANDA ()          | 42 × 8 ft  | 336 sqft | East (road)          | 5 wooden pillars, swing, sitting area, shoe rack             |

Vastu Compliance (v4 — Verified):

- Main door: East (facing road) — ideal
- NE corner: Hall + Pooja cabinet — (God's corner)
- SE corner: Kitchen — (Agni, fire element)
- SW corner: Master Bedroom — (earth, stability, owner's rest)
- NW corner: Guest Bedroom — (air, guest area)
- Baths: Central service strip, drains between rooms — correct
- Guest WC: NE veranda corner — acceptable (small, away from pooja)
- Staircase: Center-East — clockwise ascending

Bhagamandala Climate Design (Rainfall 5,000+ mm/year):

- 4 ft plinth: Mandatory. Prevents monsoon flooding. Kadappa cladding.
- 4-5 ft roof overhang: Keeps rain off walls and windows.
- Laterite walls (12"/9"): Natural humidity regulation.
- Cross ventilation: Every room has windows on 2 walls minimum.
- Bathroom waterproofing: PU membrane + Fosroc/Dr.Fixit on all wet areas.

- Anti-termite: Full treatment under foundation (essential in humid W. Ghats).
- PVC rain gutters: All roof edges → connected to RWH sump (5,000L).

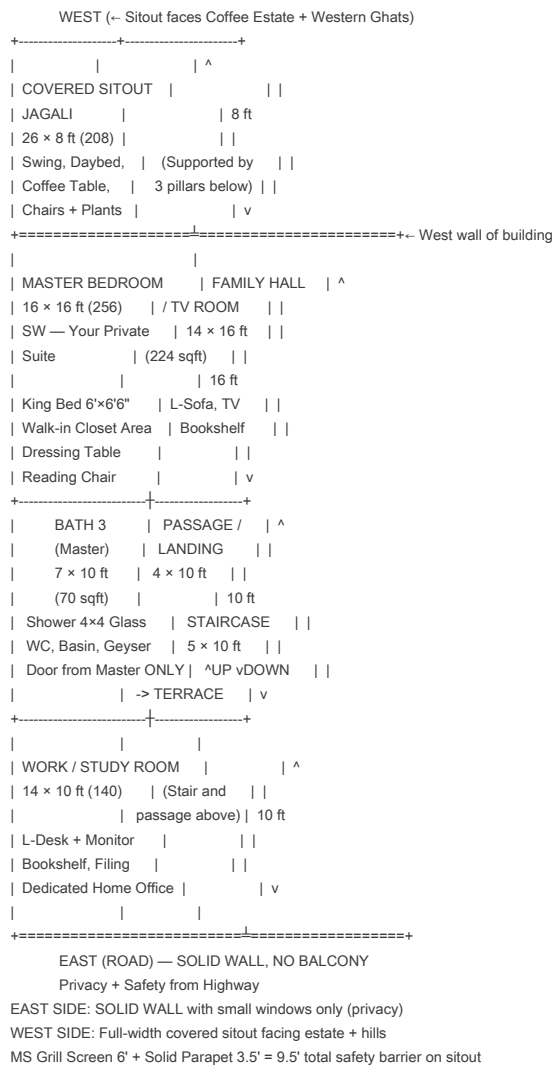
3. FIRST FLOOR PLAN (v4 — Master Suite + Work Room)

FF sits above: South row (16 ft) + Service strip (10 ft) = 26 ft N-S

FF width: 30 ft E-W (same as GF interior)

Sitout: 26 × 8 ft on WEST side (plantation + hill views)

East side: SOLID WALL (privacy + safety from highway)



First Floor Room Schedule (v4)

| Room | Size | Area | Position | Notes |
|------|------|------|----------|-------|
|------|------|------|----------|-------|

|                       |            |          |                      |                                                                          |
|-----------------------|------------|----------|----------------------|--------------------------------------------------------------------------|
| FF MASTER BEDROOM (SV | 16 × 16 ft | 256 sqft | SW — ವೈಶ್ವವರ್ತಿಯ     | King bed, walk-in closet, dressing table. Your private suite.            |
| BATH 3 (Master)       | 7 × 10 ft  | 70 sqft  | Inside service strip | Shower 4×4 glass encl., WC, basin counter, geyser. From Master ONLY      |
| FAMILY HALL / TV      | 14 × 16 ft | 224 sqft | SE area              | L-sofa, TV unit, bookshelf. Solid east wall for road privacy.            |
| WORK / STUDY          | 14 × 10 ft | 140 sqft | NE area              | Dedicated home office. L-desk, bookshelf, filing + printer.              |
| STAIRCASE             | 5 × 10 ft  | 50 sqft  | Center-East          | Continues from GF + goes to terrace                                      |
| PASSAGE / LANDING     | 4 × 10 ft  | 40 sqft  | Center               | Connects stair to all FF rooms                                           |
| COVERED SITOUT        | 26 × 8 ft  | 208 sqft | WEST side            | Swing, daybed, chairs. Plantation + hill views. 3 support pillars below. |

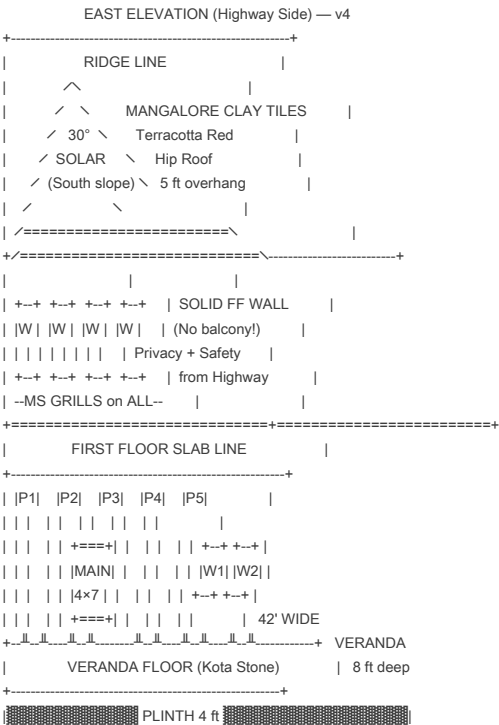
FF Safety Features (Sitout):

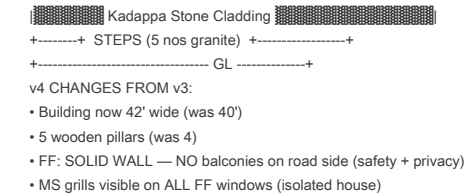
- 3.5 ft solid parapet wall (base barrier)
- 6 ft MS grill screen above parapet (total 9.5 ft barrier)
- Vertical grill bars 12mm at 100mm spacing (child-safe)
- Matt black powder coating on all grills
- Roof overhang 5 ft covers sitout from rain

Vastu (First Floor):

- Master BR: SW — stability, earth element
- Family Hall: SE — light, open
- Work Room: NE — knowledge, concentration
- Bath 3: Central, drains towards center stack
- Staircase: Continues from GF position

4. FRONT ELEVATION (EAST — ROAD SIDE)





### Front Elevation Notes:

- Ground floor: 5 Chadara Kamba style wooden pillars (wider building)
- First floor: SOLID WALL — no balcony on road side. Only windows with MS grills.
- Privacy: Highway traffic should NOT see into house
- Safety: MS grills 12mm bars on all windows. Solid wall deters intrusion.
- Plinth: 4 ft Kadappa stone cladding gives commanding presence
- Steps: 5 granite steps (each 8" rise x 10" tread)

## 5. BACK ELEVATION (WEST — ESTATE SIDE)

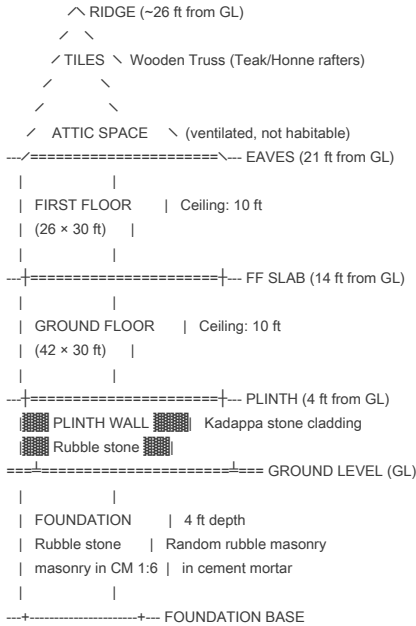


### Back Elevation Notes (v4):

- v3 had: Two separate balconies (14x6 + 12x6)
- v4 replaces with: ONE continuous covered sitout (26x8 = 208 sqft)
- Why: More usable, traditional Kodagu jagali feel, better views
- Safety: 3.5 ft solid parapet + 6 ft MS grill screen = 9.5 ft total barrier
- Support: 3 reinforced concrete pillars below (GF level = covered outdoor area)
- Partial Terrace: 12x10 ft visible near roof, accessed via staircase

## 6. SECTION DRAWING (North-South Cross Section)

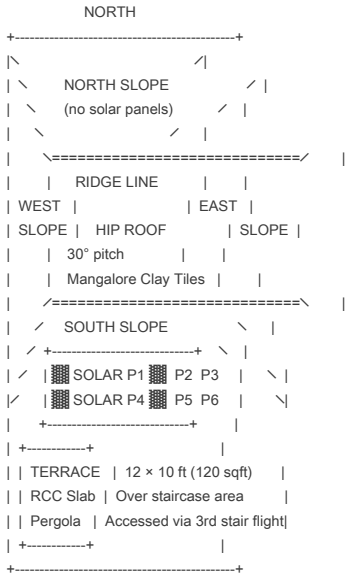
SECTION A-A' (Looking East)



## HEIGHTS:

GL to Plinth top: 4.0 ft (1.2 m)  
Plinth to GF ceiling: 10.0 ft (3.0 m)  
GF slab thickness: 0.5 ft (150 mm)  
FF floor to ceiling: 10.0 ft (3.0 m)  
FF slab thickness: 0.5 ft (150 mm)  
Eaves height: 21.0 ft from GL  
Ridge height: ~26.0 ft from GL (30° pitch)  
Terrace floor: ~25.0 ft from GL

## 7. ROOF PLAN WITH SOLAR PANEL LAYOUT



## SOUTH

SOLAR: 3 kW | 6 x 500W Mono | South slope | Grid-tied  
TERRACE: 12x10' partial | Waterproofed | Pergola | MS railing  
GUTTERS: PVC all edges -> RWH sump 5,000L

### Solar Specification:

- Capacity: 3 kW grid-tied (expandable to 5 kW)
- Panels: 6 × 500W monocrystalline (Tata/Adani/Loom Solar)
- Placement: South-facing slope (max sun)
- Inverter: 3 kW hybrid (Growatt/Havells), located in Store room
- Karnataka subsidy: Up to 40% for residential ≤ 3 kW
- Net metering: CESC.COM. Saves 15-20K/year.

Partial Terrace (12 × 10 ft):

- Slab: RCC M20, 5" thick, 5-layer waterproofing (PU membrane)
- Access: 3rd staircase flight (14 granite steps)
- Railing: 3.5 ft MS railing, matt black
- Canopy: Polycarbonate pergola (rain protection)
- Use: Views, coffee drying, solar panel access, dry-season yoga

8. POOJA UNIT DETAIL

Ready-made wall-mounted pooja cabinet on NE corner wall of Hall (GF).

- Type: Ready-made wooden pooja cabinet (wall-mounted or floor-standing)
- Size: ~3 ft wide × 1.5 ft deep × 5 ft tall
- Material: Solid sheesham or teak finish, carved border, brass fittings
- Position: NE corner of Hall — (most auspicious)
- Shelves: 3 tiers — top (deities), middle (lamp/incense), bottom (storage)
- Lighting: Brass hanging lamp hook + LED strip inside
- Cost: 8,000–15,000 (ready-made)

9. BATHROOM DETAILS (v4 — All 7 × 10 ft = 70 sqft)

All three bathrooms are now 7 × 10 ft = 70 sqft — spacious with separate shower enclosure, commode, basin counter, and geyser. Back-to-back layout on GF means single plumbing stack.

Bath Layout (applies to all 3 baths):



COMPARED TO v3:  
v3: Bath1 = 8×6 (48sqft), Bath2 = 7×5 (35sqft!) — cramped  
v4: ALL baths = 7×10 (70sqft) — LUXURY! Separate shower encl.

Bathroom Specifications (all 3):



| Item          | Specification                                    |
|---------------|--------------------------------------------------|
| Commode       | White, EWC, soft-close seat (Parryware/Hindware) |
| Basin         | Counter-top oval basin, 5' granite counter       |
| Shower        | 4×3' glass enclosure + rain shower + hand shower |
| Geyser        | 25L storage (Racold/AO Smith)                    |
| Tiles         | Anti-skid floor + wall tiles to 7' height        |
| Waterproofing | PU membrane (Dr.Fixit/Fosroc) under tiles        |
| Ventilator    | 2×1.5 ft louver at 6' height                     |
| Door          | WPC 2.5×6.5, SS tower bolt + indicator lock      |

10. DOOR SCHEDULE (v4)

| #   | Location                  | Size (W×H)   | Material     | Type                | Hardware                          |
|-----|---------------------------|--------------|--------------|---------------------|-----------------------------------|
| D1  | Main entrance (East)      | 4 × 7 ft     | Burma Teak   | Double leaf, carved | Brass mortise lock, brass handles |
| D2  | Master Bedroom (GF)       | 3 × 7 ft     | Country wood | Single leaf, panel  | SS mortise lock, tower bolt       |
| D3  | Bedroom 2 (GF)            | 3 × 7 ft     | Country wood | Single leaf, panel  | SS mortise lock, tower bolt       |
| D4  | Kitchen (from veranda)    | 3 × 7 ft     | Country wood | Single leaf, panel  | SS handle, magnetic catch         |
| D5  | Kitchen rear door         | 2.5 × 6.5 ft | Country wood | Single leaf         | SS handle, latch                  |
| D6  | Bath 1 (inside Master)    | 2.5 × 6.5 ft | WPC          | Single leaf, flush  | SS tower bolt + indicator lock    |
| D7  | Bath 2 (inside BR2)       | 2.5 × 6.5 ft | WPC          | Single leaf, flush  | SS tower bolt + indicator lock    |
| D8  | Store (from kitchen)      | 2.5 × 6.5 ft | Country wood | Single leaf, flush  | Pad bolt                          |
| D9  | Guest WC (veranda)        | 2 × 6.5 ft   | WPC          | Single leaf, flush  | SS tower bolt + indicator lock    |
| D10 | FF Master BR              | 3.5 × 7 ft   | Country wood | Carved panel        | SS mortise lock, tower bolt       |
| D11 | Bath 3 (inside FF Master) | 2.5 × 6.5 ft | WPC          | Single leaf, flush  | SS tower bolt + indicator lock    |
| D12 | Family Hall (FF)          | 3 × 7 ft     | Country wood | Single leaf, panel  | SS mortise lock                   |
| D13 | Work Room (FF)            | 3 × 7 ft     | Country wood | Single leaf, panel  | SS mortise lock                   |
| D14 | Sitout French Door        | 4 × 7 ft     | Country wood | Double leaf, glass  | SS handle, tower bolt             |

11. WINDOW SCHEDULE (v4 — MS Grills on ALL windows)

| #  | Location     | Size (W×H) | Sill Ht | Type                         | Notes                   |
|----|--------------|------------|---------|------------------------------|-------------------------|
| W1 | Hall (East)  | 5 × 4 ft   | 3 ft    | Double casement + ventilator | MS grill, mosquito mesh |
| W2 | Hall (East)  | 5 × 4 ft   | 3 ft    | Double casement + ventilator | MS grill, mosquito mesh |
| W3 | Hall (North) | 4 × 4 ft   | 3 ft    | Double casement              | MS grill, mosquito mesh |

|        |                           |            |      |                              |                             |
|--------|---------------------------|------------|------|------------------------------|-----------------------------|
| W4     | Kitchen (East)            | 4 × 4 ft   | 3 ft | Double casement              | MS grill, mosquito mesh     |
| W5     | Kitchen (South)           | 4 × 4 ft   | 3 ft | Single casement              | MS grill                    |
| W6     | Master BR (West)          | 5 × 4 ft   | 3 ft | Double casement + ventilator | MS grill, estate views      |
| W7     | Master BR (South)         | 5 × 4 ft   | 3 ft | Double casement              | MS grill                    |
| W8     | BR2 (West)                | 5 × 4 ft   | 3 ft | Double casement + ventilator | MS grill, estate views      |
| W9     | BR2 (North)               | 5 × 4 ft   | 3 ft | Double casement              | MS grill                    |
| W10    | Store (West)              | 3 × 3 ft   | 4 ft | Single casement              | MS grill                    |
| W11-14 | FF (East, various)        | 4 × 4 ft   | 3 ft | Double casement              | MS grill, solid wall around |
| W15    | FF Master (West → sitout) | 5 × 4 ft   | 3 ft | Double casement              | Views through sitout grill  |
| W16    | FF Master (South)         | 5 × 4 ft   | 3 ft | Double casement              | MS grill                    |
| W17    | FF Family Hall (East)     | 5 × 4 ft   | 3 ft | Double casement              | MS grill, highway side      |
| W18    | FF Family Hall (South)    | 4 × 4 ft   | 3 ft | Single casement              | MS grill                    |
| W19    | FF Work Room (East)       | 4 × 4 ft   | 3 ft | Double casement              | MS grill, daylight for work |
| W20    | FF Work Room (North)      | 4 × 4 ft   | 3 ft | Single casement              | MS grill                    |
| V1-V4  | All bathrooms + GWC       | 2 × 1.5 ft | 6 ft | Louver ventilator            | MS grill                    |

All windows: 12mm MS grill bars at 100mm spacing. Wooden frame, glass panes + wooden shutters. Mosquito mesh on bedrooms & hall.

**SAFETY NOTE: MS grills on ALL windows are essential — isolated house on highway curve, no neighbors nearby.**

## 12. COMPLETE COLOR PALETTE & FINISHES

### Exterior

| Element            | Color                  | Code / Material                | Finish                          |
|--------------------|------------------------|--------------------------------|---------------------------------|
| Walls              | Warm Cream / Off-white | Asian Paints Ivory Tusk 8019   | Exterior emulsion, weatherproof |
| Plinth             | Natural Grey           | Kadappa stone cladding         | Unpolished, natural             |
| Woodwork           | Dark Teak Brown        | Wood stain + PU marine varnish | Semi-gloss, UV resistant        |
| Roof tiles         | Terracotta Red         | Mangalore clay tiles (natural) | Unglazed                        |
| MS Railings/Grills | Matt Black             | Powder coating                 | Matte                           |
| Steps              | Grey Granite           | Sadarahalli granite            | Flame-finished, non-slip        |
| Compound wall      | Cream                  | AP Ivory Tusk 8019             | Exterior emulsion               |

### Interior

| Room               | Wall Finish              | Material / Code     | Ceiling      | Flooring                      |
|--------------------|--------------------------|---------------------|--------------|-------------------------------|
| Hall / Living (GF) | Light Warm White (paint) | AP 8147 Marshmallow | Bright White | Beige wood-look vitrified 2×4 |

|                     |                                |                        |                    |                           |
|---------------------|--------------------------------|------------------------|--------------------|---------------------------|
| Kitchen             | Butter Cream (paint)           | AP 7958 Buttercup      | Bright White       | Anti-skid ceramic         |
| Master Bedroom (GF) | Exposed Laterite + Linseed Oil | — NO plaster, NO paint | White Matte        | Beige wood-look vitrified |
| Bedroom 2 (GF)      | Exposed Laterite + Linseed Oil | — NO plaster, NO paint | White Matte        | Beige wood-look vitrified |
| FF Master BR        | Exposed Laterite + Linseed Oil | — NO plaster, NO paint | White Matte        | Beige wood-look vitrified |
| Family Hall (FF)    | Light Warm White (paint)       | AP 8147 Marshmallow    | White Matte        | Beige wood-look vitrified |
| Work Room (FF)      | Warm Orange tint (paint)       | AP Ivory Spark 8013    | White Matte        | Beige wood-look vitrified |
| All Bathrooms       | White tiles to 7'              | —                      | White cement paint | Anti-skid ceramic         |
| Passage / Staircase | White (paint)                  | AP Pure White          | White Matte        | Vitrified / Granite steps |

Exposed Laterite Wall Treatment — Kodagu Ainmane Heritage

All three bedrooms (Master GF, BR2 GF, FF Master) feature exposed laterite stone walls ( ) with linseed oil finish — a heritage design choice honoring the Kodagu Gowda Ainmane tradition.

Why Exposed Laterite in Bedrooms:

- Heritage: Traditional Kodagu Ainmane (ancestral homes) always used exposed laterite — warm, earthy, nostalgic
- Durability: No repainting every 3-5 years. Laterite lasts generations with minimal maintenance
- Climate: Laterite breathes — absorbs/releases humidity naturally (ideal for 5,000mm/year rainfall zone)
- Cost: Saves ~2.5–3L by eliminating plaster + paint on ~1,200 sqft of bedroom wall area
- Thermal: Natural insulation — cool in summer, warm in winter

Linseed Oil Treatment Process:

1. After masonry cures (28 days), brush clean all dust from laterite surface
2. Apply raw linseed oil (Alsi Tel / ಅಣ್ಣೆ ಎಣ್ಣೆ) with wide brush — 2 coats, 24hr gap
3. Final coat: boiled linseed oil for faster drying and richer sheen
4. Re-apply once every 3-5 years (vs repainting which costs 30-50K each time)
5. Result: Warm terracotta-red satin finish that deepens with age

Which walls are exposed laterite:

- All interior bedroom walls (4 walls per room × 3 rooms)
- Ceiling remains white matte paint (plastered RCC slab)
- Other rooms (hall, kitchen, baths, passage) remain plastered + painted as specified above

13. EXTERNAL WORKS

Garden & Landscaping

- Front (East): Interlock driveway, pathway to veranda, 2 coconut trees flanking gate, low Ixora hedge
- NE corner: Tulsi Vrindavan (raised stone platform)
- Sides: 3-5 ft gap with PCC drain along foundation. Arecanut palms (north).
- Back (West): Small lawn, jasmine creeper on wall, bougainvillea at corners
- Coffee Drying Yard (30×20 ft): PCC floor, retractable polycarbonate roof, behind parking

Compound Wall & Gate (v4 — SAFETY PRIORITY)

+----- COMPOUND WALL -----+

|  
| ROAD SIDE (East): |  
| 7 ft SOLID laterite masonry — NO railing on top |  
| Crash protection from highway curve traffic |  
| Reflectors embedded for night visibility |  
| Plastered + cream paint |  
|  
| OTHER SIDES (N, W, S): |  
| 6 ft SOLID laterite — NO railing (security) |  
| Anti-climb (smooth plaster on top) |  
|  
| MAIN GATE (East): |  
| 12 ft vehicle + 4 ft pedestrian |  
| MS, matt black, SET BACK from curve apex |  
| 2 laterite pillars with motion-sensor LED lights |  
|  
| CCTV: 4 cameras (gate, road curve, back, parking) |  
| MOTION LIGHTS: 2 front + 2 back LED floods |  
|  
+-----+

14. ELECTRICAL LAYOUT (v4 — includes Work Room)

| Room             | Lights            | Fans      | 5A | 15A               | Switches    | Special                              |
|------------------|-------------------|-----------|----|-------------------|-------------|--------------------------------------|
| Hall/Living      | 2 (LED panel)     | 1         | 4  | 1 (AC)            | 6           | TV point, USB charging               |
| Kitchen          | 2 (LED panel)     | 1         | 4  | 2 (fridge, mixer) | 6           | Exhaust fan, chimney point           |
| Master BR (GF)   | 1 (LED panel)     | 1         | 4  | 1 (AC)            | 4           | Bedside USB (2 nos)                  |
| Bedroom 2        | 1 (LED panel)     | 1         | 4  | 1 (AC)            | 4           | Study lamp point                     |
| Bath 1           | 1 (LED mirror)    | 1 exhaust | 1  | 1 (geyser)        | 2           | —                                    |
| Bath 2           | 1 (LED mirror)    | 1 exhaust | 1  | 1 (geyser)        | 2           | —                                    |
| Store            | 1 (LED tube)      | —         | 1  | 1 (inverter)      | 1           | Solar inverter + DB                  |
| Staircase        | 2 (LED wall)      | —         | —  | —                 | 2 (two-way) | Motion sensor                        |
| Veranda          | 2 (LED wall+ceil) | 1         | 2  | —                 | 3           | Gate bell, CCTV point                |
| FF Master BR     | 1 (LED panel)     | 1         | 4  | 1 (AC)            | 4           | Bedside USB (2 nos)                  |
| Bath 3           | 1 (LED mirror)    | 1 exhaust | 1  | 1 (geyser)        | 2           | —                                    |
| Family Hall (FF) | 2 (LED panel)     | 1         | 4  | 1 (AC)            | 6           | TV point                             |
| Work Room (FF)   | 2 (LED panel)     | 1         | 6  | 2 (PC, printer)   | 6           | LAN port, 2× USB-C, UPS point        |
| Sitout           | 2 (LED wall)      | —         | 1  | —                 | 2           | Waterproof socket                    |
| External         | 4 (motion LED)    | —         | —  | —                 | 2           | CCTV power (4 pts)                   |
| Ext. Taps        | —                 | —         | —  | —                 | —           | 2 external bib cocks                 |
| Washing Machine  | —                 | —         | —  | 1 (15A)           | 1           | Kitchen NE, cold water inlet + drain |

Total Load: ~6 kW (no AC) / ~12 kW (with ACs)  
Supply: 3-phase CESCO connection  
Solar: 3 kW grid-tied via net meter  
Work Room: Dedicated 15A circuit for computer equipment + UPS socket

15. PLUMBING LAYOUT (v4 — Single Stack Advantage)

v4 Plumbing Advantage: BACK-TO-BACK BATH CORE

- Bath 1 and Bath 2 share a wall → single plumbing stack for both GF bathrooms:
- ONE main soil pipe (4" PVC)
  - ONE common vent pipe
  - Shared hot water riser from solar heater
  - Saves 30-40K in plumbing cost vs separate stacks

Water Supply

- Source: Open/bore well (West side)
- Pump: 1 HP submersible
- Overhead tank: 2,000L syntax on MS stool above staircase
- Solar water heater: 200L on FF slab (South)
- Hot water: Solar to all 3 baths via single riser. Electric geyser backup each bath.
- RWH: Roof gutters → first flush → 5,000L sump (NW)

Drainage

- Greywater: PVC → Soak pit (NW)
- Blackwater: PVC → Septic tank (NW, 2-chamber, 15 ft from well)
- Kitchen waste: Grease trap → Grey water line
- Rainwater: Open drains → Recharge pit → Natural slope

External Water Taps (IMPORTANT — mark during plumbing rough-in)

- Tap 1 — Entrance (Right of main steps): 15mm GI pipe from main line. Chrome bib cock. Drain plate below → connects to front garden drain. Purpose: Wash feet before entering house (Kodagu tradition, also muddy monsoon months).
- Tap 2 — Parking/Garage (Left side, near garage): 15mm GI pipe from main line. Chrome bib cock + hose connector. Drain to garden slope. Purpose: Gardening watering, car wash, general outdoor cleaning.
- Both taps: Run pipes during wall construction stage itself. Embed in compound wall / plinth wall. Leave stub-out with cap until fixture stage.
- Note: Both external taps must have individual stop cocks (gate valve) inside the store room near the main line for isolation.

Washing Machine Point (Kitchen)

- Location: Kitchen NE corner, near washing area (x=14ft, z=20ft in plan)
- Water: Cold water inlet with wall-mounted bib cock (15mm)
- Drain: Floor drain 3" connected to greywater line
- Electrical: Dedicated 15A socket, 1.2m height from floor
- Note: Leave space for front-load washing machine 24"×24"×33"

16. AREA STATEMENT (v4 — FINAL)

House Only

| Component | Sqft | Notes |
|-----------|------|-------|
|-----------|------|-------|

|                                 |             |                                  |
|---------------------------------|-------------|----------------------------------|
| GROUND FLOOR                    |             |                                  |
| Hall / Living (NE)              | 256         | 16 × 16, full rectangle          |
| Master Bedroom (SW)             | 256         | 16 × 16, full rectangle          |
| Bedroom 2 (NW)                  | 224         | 14 × 16, full rectangle          |
| Kitchen (SE)                    | 224         | 14 × 16, full rectangle          |
| Bath 1 (Master)                 | 70          | 7 × 10, in service strip         |
| Bath 2 (BR2)                    | 70          | 7 × 10, in service strip         |
| Store                           | 80          | 8 × 10, adj. kitchen             |
| Staircase                       | 50          | 5 × 10                           |
| Passage                         | 30          | 3 × 10                           |
| Guest WC                        | 16          | 4 × 4, veranda corner            |
| Walls (~10%)                    | ~130        |                                  |
| GF Enclosed Total               | ~1,406 sqft |                                  |
| Veranda (42 × 8, net after GWC) | 320         | Covered                          |
| GF Total Built-up               | ~1,726 sqft |                                  |
| FIRST FLOOR                     |             |                                  |
| FF Master Bedroom (SW)          | 256         | 16 × 16                          |
| Bath 3 (Master)                 | 70          | 7 × 10                           |
| Family Hall / TV                | 224         | 14 × 16                          |
| Work / Study Room               | 140         | 14 × 10                          |
| Staircase                       | 50          | 5 × 10                           |
| Passage / Landing               | 40          | 4 × 10                           |
| Walls (~10%)                    | ~75         |                                  |
| FF Enclosed Total               | ~855 sqft   |                                  |
| Covered Sitout (26 × 8)         | 208         | West side                        |
| FF Total Built-up               | ~1,063 sqft |                                  |
| Partial Terrace                 | 120         | 12 × 10, RCC slab                |
| HOUSE TOTAL BUILT-UP            | ~2,909 sqft | incl. veranda + sitout + terrace |
| HOUSE CARPET AREA               | ~2,261 sqft | enclosed rooms only (GF+FF)      |

External Structures

|                              |            |
|------------------------------|------------|
| Structure                    | Sqft       |
| Open Garage (15 × 25)        | 375        |
| Tool Shed (10 × 8)           | 80         |
| Coffee Drying Yard (30 × 20) | 600        |
| External Total               | 1,055 sqft |

v4 vs v3 Comparison:

|                    |              |                |               |
|--------------------|--------------|----------------|---------------|
| Metric             | v3           | v4             | Change        |
| Building footprint | 40×32        | 42×38          | +216 sqft     |
| GF enclosed        | ~1,192       | ~1,406         | +214 sqft     |
| FF enclosed        | ~570         | ~855           | +285 sqft     |
| Total built-up     | ~2,222       | ~2,909         | +687 sqft     |
| Bedrooms           | 3 (L-shaped) | 3 (RECTANGLES) | Better layout |

|                 |                    |                          |              |
|-----------------|--------------------|--------------------------|--------------|
| Bath sizes      | 48/35/48 sqft      | 70/70/70 sqft            | +75% avg     |
| Balconies       | 2 small (156 sqft) | 1 full sitout (208 sqft) | +33%         |
| Work room       | None               | 140 sqft                 | NEW          |
| Plumbing stacks | 3 separate         | 1 shared (GF)            | Saves 30-40K |

## 17. COST ESTIMATE

Cost details are maintained in a separate private document.

Contact the owner for cost-related discussions.

## 18. IMPORTANT NOTES FOR WORKERS / CONTRACTORS

### Kannada Summary ( )

ಮನೆ ನಕ್ಷೆಯು (v4):

- ಒಟ್ಟು ಮನೆ ವಿಸ್ತೀರ್ಣ: ~2,900+ ಚ.ಅ. (G+1)
- ಕಟ್ಟಡ ಅಳತೆ: 42 x 38 ಅಡ (ಒಳಗಡೆ ಸೇರಿದ)
- ಅಡ್ಡದಿ: 4 ಅಡ ಅಳ, ರಬ್ಬರ್ ಕಲ್ಲು ಗೋಡೆ
- ಪಾಲಿಂಗ್: 4 ಅಡ ಏಕತರ (ಮೇಲೆ ಪರದಾಳ - ಆತ್ಮೀಯ ಮುಖ್ಯ)
- ಗೋಡೆ: 12" ಲೇಟೈಟ್ (ಹೊರಗಡೆ), 9" ಲೇಟೈಟ್ (ಒಳಗಡೆ)
- ಮಾಡು: ಮರದ ಟ್ರಸಸ್ (ಹೊನ್ನೆ/ತೇಕ), ಮಂಗಲೂರು ಹಂಚು, 30° ಇಳಿಜಾರು
- RCC ಸ್ಲಾಬ್: FF + ಸ್ಲಾಬ್ + ಟೆರೇಸ್ (5" ದಪ್ಪ)
- ಸೇವಾನ್ವಯ: ಎಲೆಕ್ಟ್ರಿಕ್ 3 = 7x10 ಅಡ (70 ಚ.ಅ. — ವಾತಾವರಣ)
- ಎಲೆಕ್ಟ್ರಿಕ್ ಕ್ಯಾಪ್: MS ಗೆರಾಲ್ (ಫರ್ಲಾ — ಮುಖ್ಯಯರಸಾತ್ ಬಳಸಿ)
- ಕಾಂಪೌಂಡ್: 7 ಅಡ ಗೋಡೆ (ರಸಾತ್ ಕಡೆ), CCTV 4 ಕೇಯಮೂರು

### Critical Construction Sequence

- Foundation: Excavate 4 ft → PCC bed → Rubble stone masonry → Anti-termite treatment
- Plinth: 4 ft height MANDATORY. Kadappa cladding.
- GF walls: 12" laterite external, 9" internal. Build service strip walls carefully — back-to-back baths need precise alignment for plumbing stack.
- RCC slab: M20 grade, 5" thick, Fe500 steel. CURE MINIMUM 14 DAYS. Note: Sitout cantilever/pillars need structural engineer's calculations.
- FF walls: Same as ground floor.
- Roof truss: Honne/Teak, 30° pitch. Extended 5 ft to cover sitout on west.
- Mangalore tiles: Hip roof with ridge tiles. 5 ft overhang all sides.
- Waterproofing: Critical for Bhagamandala. All baths, sitout, terrace, FF slab.
- Plastering → Electrical → Plumbing → Flooring → Painting → Woodwork

### Waterproofing (CRITICAL — 5,000+ mm/year rainfall)

- ALL bathroom floors: PU membrane + waterproof tile adhesive
- FF slab top: Bituminous waterproofing before screeding
- Sitout floor: Waterproof membrane + 1.5% slope towards drain
- Terrace: 5-layer waterproofing (primer, PU, screed, tiles, sealant)
- External walls: 2 coats waterproof cement paint
- Plinth: DPC bitumen layer at plinth level

## Safety Features (ISOLATED HIGHWAY LOCATION)

- 7 ft solid compound wall on road side (crash + security)
- 6 ft solid walls on all other sides (no MS railing — full solid)
- MS grills on ALL windows (12mm bars, 100mm spacing)
- CCTV 4 cameras with DVR in work room
- Motion sensor LED lights (front gate area + back)
- Reflectors on compound wall (highway curve — night safety)
- Gate set back from curve apex (vehicle turning clearance)
- **Sitout: 3.5ft parapet + 6ft MS grill = 9.5ft barrier**
- **Main door: Double deadbolt + latch**

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Prepared by: Owner with AI assistance

Date: April 2026 (v4 — Optimized Layout)

Location: Bhagamandala–Talakaveri Road, Kodagu

Note: This plan requires structural engineer's approval and Gram Panchayat building permission before construction.